

PDEUE & IFAS SYSTEM ARCHITECTURE SPECIFICATION

Reconciled Governance, Role-Based Desktops, Capital Flow & Accounting Boundary | Binding Lexicon v0.4

1. CHIEF ADMINISTRATOR COCKPIT & GRANULAR ANALYTICS ENGINE

The Chief Administrator Workspace (/dashboard) serves as the primary governance center for the Financial Generation System. It integrates forensic trade drill-down, live order book microstructures, and multi-timeframe capital tracking.

Granular Contract Drill-Down (IF-015) Enables point-in-time forensic inspection. Clicking any active or historical contract slides out a detailed telemetry drawer displaying raw NWP weather ensemble vectors, BLS CPI release components, model probability curves, and VWAP execution timestamps.	Microstructure & Slippage Barrier Visualizes resting Central Limit Order Book (CLOB) depth across Kalshi and Polymarket. Tracks dynamic order-book walking and fail-closed aborts whenever market sweeps induce slippage exceeding the immutable 1.5% ceiling.
Multi-Timeframe Compounding Curve Interactive equity visualizer aggregating performance across 1M, 1H, 12H, RTH, 24H, 7D, 14D, 1MO, 1Q, and 1Y horizons. Compares uncommitted cash against mark-to-market valuations and realized waterfall compounding.	Componentized Template Library Standardized CSS/HTML UI widgets (cards, compounding lines, depth step-charts, and pure-CSS formula tooltips) designed for zero-code reusability across Member and Advisor sub-portals.

2. ROLE-BASED DESKTOP SUITE & ONBOARDING PROVISIONING

To maintain least-privilege security and prevent cross-tenant contamination, PDEUE partitions user interfaces into segregated operational domains:

Portal & Route	Authorized Role	Functional Scope & Feature Surface	Hard Safety Boundaries
Chief Admin /dashboard	Chief Administrator / Founder	Master engine mode toggle, emergency kill switch, dual-control approval queues, CFCP/FAEP meters, member provisioning modal.	Requires Hardware MFA / WebAuthn. State changes generate structured Action Cards.
Member Portal /member	Lineal Family Members (SCMAs)	Single-SCMA balance, personal compounding curve, downward-only risk dial (0.5%-5%), autonomous distribution requests, Academy.	Strictly isolated. Cannot view peer balances, exchange keys, or system-wide risk settings.
Advisor Desk /advisor	Lineal Advisors (Tiers F1-F3)	Household lineage risk analytics, multi-SCMA exposure aggregation, plain-English decision packet audits, Red-Tier co-signing.	Read-only / advisory posture. No trade execution, risk dial mutation, or kill-switch controls.
System Console /admin/tech	Technical Operators (T1-T3)	Daemon polling health, token-bucket rates, container CPU/RAM loads, exchange WebSocket heartbeat diagnostics.	Redacted financial plane. Cannot view dollar balances, member names, or ledger books.

3. FINANCIAL ACCOUNTS, CASH FLOWS & DECENTRALIZED DISTRIBUTIONS

PDEUE enforces strict legal and structural separation between physical money custody and virtual trading state. The engine operates as a Financial Generation Engine (FGE) rather than an authoritative bank depository.

- **Master Custodial Account (FBO Entity):** Commercial checking or trust account opened under the family investment LLC/Trust 'For Benefit Of' family members. Holds seed principal. PDEUE has zero direct credentials or banking API connections.
- **Venue Escrow Collateral:** CFTC-regulated USD escrow at Kalshi and USDC smart-contract proxy at Polymarket. Mapped internally as exact integer cents.
- **Member Personal Bank Accounts:** External retail accounts owned by members. Linked to PDEUE exclusively via non-sensitive reference tokens (e.g., EXT-REF-CHASE-****4812). Real account/routing numbers are never stored in PDEUE.
- **Virtual Double-Entry Ledgers (ADR-008):** 64-bit integer-cent sub-ledgers partitioning cash across Self-Contained Member Accounts (SCMAs), the 10% Central Familial Common Pool (CFCP), and the 3% Founder Endowment Pool (FAEP).

- **Platform Yield Accrual:** The PlatformYieldAdapter captures Treasury bill yields paid by venues on unallocated cash, crediting interest back to originating member SCMAs or common reserves.

Decentralized Member Distribution Gateway:

GREEN TIER (Routine Profit)	YELLOW TIER (Buffer Advisory)	RED TIER (Capital Preservation)
Threshold: Net profit withdrawal leaving original seed and operating buffer intact. Flow: Instant, autonomous execution. No human gatekeepers. Emits signed ADR-009 event directly to IFAS for bank ACH.	Threshold: Withdrawal dips into operational buffer or reduces compounding velocity. Flow: AI Copilot generates educational impact card. Member reviews velocity trade-off and retains final execution authority.	Threshold: Requests drawing down seed principal or rapid successive drawdowns. Flow: 24-hour cooling-off period. Requires dual-confirmation from assigned F1/F2 mentors to prevent panic liquidation.

4. LINEAL MENTORSHIP, GOVERNANCE & DELEGATION

- **F1 / F2 Mentorship Layer:** Every junior member is paired with a peer mentor (F1) and household senior lead (F2) to build generational financial literacy. Mentors review progress and co-sign Red-Tier withdrawals without holding trading authority.
- **House-Tabulated CFCP Treasury Governance:** To prevent large family branches from dominating common treasury allocations, CFCP grant proposals require a bicameral threshold: a member popular vote (>50%) AND a supermajority (>=75%) of recognized Lineal Houses.
- **Executive Administrative Delegation:** The Chief Administrator delegates technical cloud operations to the CTO (T3) and financial risk bounds to the CRO (F3) via dual-control co-signatures, backed by cryptographic session leases for acting administrators.

5. INDEPENDENT FINANCIAL ACCOUNTING SYSTEM (IFAS) BOUNDARY

IFAS is a physically and logically decoupled general-ledger application operating as an asynchronous consumer of PDEUE event streams. Development commences following portal template staging and prior to production capital funding.

- **Signed Event Boundary (ADR-009):** PDEUE emits immutable, sealed JSON-LD records (BALANCE_SETTLEMENT, WATERFALL_DISTRIBUTION, CAPITAL_SWEEP) signed via HMAC-SHA256. IFAS validates signatures and returns idempotent receipts.
- **Zero Depository Credentials in Engine:** PDEUE contains zero bank routing or account credentials. All fiat wire and ACH disbursements are managed exclusively inside IFAS behind institutional banking controls.
- **Physical & Network Isolation:** Hosted in a dedicated VPC at accounting.narratix.io. Communicates with PDEUE via authenticated REST outbox APIs over TLS 1.3 with hardware security key (FIDO2) enforcement.

6. COMMUNICATIONS, EDUCATIONAL ACADEMY & NARRATIX.IO TOPOLOGY

- **Lineage Community Hub (/community):** Isolated family hearth for non-business communications, milestone celebrations, and family news, firewalled completely from trading engine databases.
- **Sandboxed Video Conferencing:** Jitsi Meet container embedded via an isolated HTML iframe (allow='camera; microphone') on meet.narratix.io with zero access to API tokens or financial state.
- **Interactive AI Academy:** AICopilotEngine delivers on-the-fly tutoring, explaining binary option geometry, Kelly sizing, and point-in-time trade decisions directly inside the UI.
- **Unified narratix.io Subdomain Routing:**

Subdomain	Target Service	Access Protocol & Purpose
admin.narratix.io	Chief Admin Cockpit	Hardware MFA / FIDO2. Mode switches, kill switch, dual control.
family.narratix.io	Member & Advisor Portals	OAuth2. Single-SCMA views, downward risk dials, Academy hub.
pdeue.narratix.io	Trading Engine Core	API Token / IP Allowlist. Market data ingestion & execution dispatch.
accounting.narratix.io	IFAS Gateway	Private Cloud / Cloudflare Tunnel. General ledger & tax export.